LESSON PLAN

COMPOSITE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	11	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Micrometer, descriptions of various parts, important terms such as backlash, measuring range, total error, cumulative error: various types
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Micrometer, descriptions of various parts, important terms such as backlash, measuring range, total error, cumulative error: various types of Micrometers.
5. Specific Objectives	19.1.1.1 Trainee will be able to describe details about Engineering Metrology.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Micrometer parts identification, explanation of backlash and error terms, Instructor supervision of model handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a micrometer?
(b) Name two other precision measuring instruments used in engineering.
(c) Why is precision measurement critical in aircraft maintenance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Micrometer frame and anvil description • Sleeve and thimble scale markings • Ratchet stop and locking mechanism • Micrometer head assembly details	L & Ds	PPT, IB	15 min
(b)	• Definition and cause of backlash • Measuring range and span concepts • Total error calculation methods • Cumulative error accumulation effects	L & Ds	PPT, PDF	20 min
(c)	• External micrometer types and uses • Internal micrometer variations • Depth micrometer applications • Selection criteria for micrometer types	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how backlash affects measurement accuracy.
(b) Differentiate between total error and cumulative error.
(c) Identify the correct micrometer type for measuring internal bores.

12. Home Assignment and Follow Up	Prepare a brief note on the care and handling of micrometers to be discussed in the next lesson on Slip Gauges.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Slip gauges introduction, Indian standards slip gauges, care and use of slip gauges	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min